Chapter 3
Statistics Applied to Metrology



3.1 Significant Digits of a Measure

The result of a calculation using all the calculator's display digits implies that it is accurate for all digits, a fact that, in practice, is rarely possible (the number of digits can be increased considerably using computers).

When we use the measurement results from calculations, we must consider that the numbers used have only a limited number of significant digits, because the concepts of uncertainty, accuracy, resolution, and conversion of units are involved.

Suppose measurement 13.403 m indicates the most likely value of a quantity, and the maximum variation in the measurement series to calculate this value is 0.04 m. As this variation can be more or less, we must express the result of the measurement as follows:

$$(13.403 \pm 0.04)\ \text{m}$$

Analyzing the result, we note that the second decimal digit of the most likely value is uncertain. Therefore, it is unnecessary to write the third decimal, since the previous one is already uncertain.

The measurement result must be expressed as (13.40 ± 0.04) m.

Of the considerations made, we can establish the concept of significant digits of a measure.

Attention!
The significant digits of a measure are the digits considered correct, from the first different from zero, plus the latter, which is regarded as the doubtful significant digit.

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digits. Ambiguity can be avoided by using exponential or "scientific" standard notation.

If the number of significant digits is three, four, or five, we could write 50,600 calories, such as: 5.06×10^4 calories (three significant digits)

5.060×10^4 calories (four significant digits)
5.0600×10^4 calories (five significant digits)

When writing a number in scientific notation, the number of significant digits is indicated by the number of numerical digits in the term "digits," as shown in the examples.

Important
The power of ten is not considered a significant digit.

3.1.1 Number Rounding

When the measure has more significant numerals than you need, we should keep only those necessary and abandon the others.

For example, measurement 34.527 m has five significant digits. If we have to express it with only three, we should write 34.5 m. If we need four, we write 34.53 m.

In the latter case, we observed that the digit of the second decimal house went from 2 to 3.

Here is the reason. If we had used 34.52 m, we would have made a mistake, for lack of it, equal to $(34.527 - 34.52)\ \text{m} = 0.007\ \text{m}$.

Using 34.53 m, we made a minor error, by excess, of:

$$(34.53–34.527)\ \text{m} = 0.003\ \text{m}.$$

When we round a number, we must keep in mind the following rules:

(a) The last digit of a number should always be added from a unit if the discarded digit is bigger than five.

Examples of rounding to three significant digits:

$$134.7\ \text{m} = 135\ \text{m}$$

$$0.03432\ \text{mm} = 0.0343\ \text{mm}$$

(b) If the discarded digit is equal to five, if there are any other numerals different from zero after the discarded five, the last retained digit will be plus a unit.

Examples of rounding to three significant digits:

$$14.751\,^{\circ}\mathrm{C} = 14.8\ ^{\circ}\ \mathrm{C} \quad 0.0346501\ \mathrm{km} = 0.0347\ \mathrm{km}$$

(c) If the discarded digit is five, if there are only zero or no other digit after five, the last retained digit will be added to a unit only if it is odd.

Examples of rounding to three significant digits:

$$4.8350\ \mathrm{N} = 4.84\ \mathrm{N}$$

$$34.25\ ^{\circ}\mathrm{C} = 34.2\ ^{\circ}\mathrm{C}$$

3.1.2 Operations with Significant Digits

We must act as follows in mathematical operations for the result of operations containing significant digits only.

3.1.2.1 Addition and Subtraction

We usually add or subtract, and the operation result must have the same number of decimal digits in the portion with *the smallest number of decimal digits.*

Example: Give the result of the sum (85.45 m + 5.6 m + 98.523 m) with the correct number of significant digits.

Solution: Add the numbers and provide the result with the decimal digits in the portion with the fewest digits.

$$\begin{array}{r} 85.45 \\ 5.6 \\ +\ 98.523 \\ \hline 189.573 \end{array}$$

As the portion with fewer decimal digits is 5.6 (one decimal), adopting the sum and subtraction rule present will result in **189.6 m**.

3.1.2.2 Multiplication and Division

We usually multiply or divide, and the result must have the same *number of significant digits as the portion with the smallest number of significant digits*.

Example: Give the result of the division of 89.1 m^2 by 5.4690 m, with the correct number of significant digits.

Solution: Divide and provide the result with the number of significant digits of the portion with the smallest number of significant digits.

$$\frac{89.1\ \text{m}^2}{5.4690\ \text{m}} = 16.29182\ \text{m}$$

Adopting the rule of multiplication and division, as 89.1 has only three significant digits, the result of the division will be **16.3 m.**

3.1.2.3 Square Root

The square root of a number with n significant digits can have significant digits at most n and at least $n - 1$.

Example: $\sqrt{25.5}\,\text{km}$

Since 25.5 km has three significant digits, we can represent the result as 5.05 or 5.0. The amount of significance used will depend on the accuracy of the calculation.

Example: $\left(\sqrt{25.5} + 4.8\right)\ \text{km} = (5.0 + 4.8)\ \text{km} = 9.8\ \text{km}.$

$$\left(\sqrt{25.5} + 4.81\right)\ \text{km} = (5.05 + 4.81)\ \text{km} = 9.86\ \text{km}.$$

3.1.3 Mixed Operations

When using a calculator, if you work all the long calculations without writing the intermediate results, you may be unable to tell if an error has been made. In addition, even if you realize there were any errors, you may be unable to say where it is. In an extensive calculation involving mixed operations, as many digits should be performed as possible in the entire set of calculations, and then the result should be appropriately rounded.

For example: (5.00/1.235) m + 3.000 m + (6.35/4.0) m = (4.04858 + 3.000 + 1.5875) m = 8.630829 m.

The first division should result in three significant digits. The last division should result in two significant digits. The three added numbers should result in a number with one digit after the decimal home. Thus, the correct rounded result must be **8.6 m**. The last operation (division) limits this result's accuracy.

Important
In an extensive calculation involving mixed operations, as many digits as possible should be performed in the entire set of calculations, and then the result should be appropriately rounded.

Most modern calculators allow you to load the results of intermediate calculations on the display by performing complex calculations. In doing so, it is possible to maintain the results of each calculation step without inserting the intermediate results (a practice that perhaps encourages rounding too early). This way, you can altogether avoid truncation errors introduced by intermediate rounding.

Using all digits in the result can be critical for many mathematical operations in statistics. Rounding intermediate results by calculating squares can seriously compromise their accuracy.

3.2 Concepts of Statistics Applied to Metrology

Here is a question: Why must we know the statistical tools to work in metrology?

It is well known that every measure performed has an associated measurement uncertainty, which, depending on the type and quality of the instrument or system used, can be small or large compared to the measurement's result.

Thus, we can say that MR = X ± U, where.

MR = measurement result.

X = measurement value (or the mean measurement) performed and.

U = uncertainty of the measurement.

In VIM—2.26, we find the following definition for measurement uncertainty: *"non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used."* The VIM also cites that *"the parameter may be, for example, a standard deviation called standard measurement uncertainty (or a specified multiple of it), or the half-width of an interval, having a stated coverage probability."*

We realize, then, that measurement uncertainty is an evaluated parameter through some statistical tools. As the result of measurement will always have an associated doubt (called the uncertainty of measurement), what is sought is to estimate the values of measure and uncertainty in the best possible way.

Measurement uncertainty will always exist and will never be eliminated, since the actual value of greatness is estimated (in practice, the value of the standard is used as a reference value). However, it is possible to define the limits within which the value of a measurement is found by considering a specific probability value using statistical techniques and analyses.

Experimental measurements are carried out based on random experiments, and random experiment means that which is influenced by non-controlled, random variables. Thus, measurements are random experiments and can be repeated indefinitely, and once repeated, we will probably obtain different results.

Random experiments are associated with a sample space or population. Population *N* is defined as all elements available for evaluation. This population can be a finite number (e.g., the total residents of a building) or infinite (e.g., the set of natural numbers). A sample n represents a portion of the population, but it should be chosen to present the characteristics and properly represent this origin population.

After this measurement process is completed, a specific value will represent an estimate of the measurement result. From the result of this sample and by attributing a certain degree of confidence, one can analyze the behavior of the measuring instrument as a whole. Conclusions obtained based on a sample or amount of data streamline the measurement process and reduce costs.

Suppose we need to determine the density of a solid (ρ). It is known that the density is the relationship between the mass of the body (m) and its volume (V), given by the expression:

$$\rho = \frac{m}{V} \tag{3.1}$$

Then, when measuring the mass of the body with the aid of a scale, we have the variable mass M as a random component since its value can be affected by the position in which we place the body on the scale plate, besides the characteristic of the scale itself not to reproduce the measured values repeatedly.

The same is valid for measuring the part's volume. It is affected by the temperature variation where its measurement is performed and by the instability of the measurement instruments.

This is why statistical analysis is fundamental to metrology. It enables data description from central trend measures, dispersal measures, and probability distribution, followed by the analysis and interpretation of the obtained results.

Before we discuss the concept of uncertainty of measurement and present its calculation methodology, it is necessary to introduce some basic statistics foundations and tools.

3.2.1 Random Variable, Random Experiment, and Sample Space

When we experiment with a measurement, we are subject to results that can be influenced by variables we do not control. For example, the variation in ambient temperature can influence the length of a metal part (dilation), or the relative humidity of the air can affect the mass of a moisture-absorbing substance. Finally, random variables in every measurement process can interfere or may interfere with the measurement result.

Thus, such experiments have random variables, regardless of being careful with the experiment, and we cannot avoid these influence variables. Our goal, then, will be to understand, quantify, and model these types of variations that we often find in measurements.

We can define a random experiment as anyone who provides different results, even taking all the precautions to perform the measurement procedure similarly. The set of all possible results of a random experiment is called the sample space of the experiment.

These random variables can be divided into two types: discreet and continuous.

3.2.1.1 Discreet Random Variable (Fig. 3.2)

One dice has x values (1, 2, 3, 4, 5, 6). The variable is discreet; it cannot assume intermediate values. When we launch a dice, we will not find, for example, values between 1 and 2 or between 4 and 5. A discreet random variable is a variable with a finite number of values. Other examples are the number of wrinkles in a car, the number of oranges in a basket, and the number of parts manufactured in one day.

3.2.1.2 Continuous Random Variable

For example, the room temperature of a laboratory (T) measured over a week is considered a continuous random variable, because it can take any value throughout the day and week. Therefore, a continuous random variable assumes infinite values. Examples include temperature measurement, pressure measurement, and electric current measurement.

Note that the typical variables of interest in metrology are continuous. The most commonly used continuous probability distributions in metrology are:

- Uniform or rectangular distribution.
- Triangular distribution.
- Normal or Gaussian distribution.
- Student's t distribution.

This chapter will study these probability distributions and their main characteristics.

Fig. 3.2 A dice



3.2.2 Distribution of Measured Data

3.2.2.1 Not Grouped Data

For example, the CEN-EN 837–1 standard states that the type Bourdon manometer's calibration temperature should be between (20 ± 2) °C. Here, we have 80 measurements performed over one day, and the results are found in Table 3.1.

Question: Are all measurements within the tolerance range established by the standard?

Answer: Yes! After analyzing all 80 data points, it is clear that the lowest value was 18 °C, and the largest was 22 ° C.

Let us ask other questions:

- What was the value that prevailed in the measurement set?
- Did measurements vary a lot or little?
- Have you ever thought that instead of 80 measurements, we had 8000?
- Do you agree that there should be a more appropriate way to dispose of this data to facilitate answers?

3.2.2.2 Grouped Data

The immediate way is to group the data in a list. The list consists of grouping this data increasingly or decreasingly. For the temperature example, we will put it in ascending order of values (Table 3.2).

The list makes it easy to answer if the results are within tolerance. Just look at the first (18 °C) and the last (22 °C), but it is not so immediate to answer the other questions!

Table 3.1 Temperature measurements

19	21	20	21	19	20	19	19
22	20	18	18	20	22	21	21
18	21	19	21	18	19	19	22
21	19	22	22	20	20	22	22
21	18	19	18	18	20	18	21
20	19	20	21	22	20	18	21
19	18	22	21	18	21	19	19
19	19	19	20	22	18	21	21
18	19	18	21	22	22	21	21
20	21	19	22	18	21	21	21

Table 3.4 Accumulated absolute frequency (F_a)

Value °C	f_a	F_a
18	15	15
19	18	33
20	12	45
21	22	67
22	13	80

Table 3.5 Simple relative frequency (f_r)

Value °C	f_a	f_r
18	15	18.75%
19	18	22.5%
20	12	15%
21	22	27.5%
22	13	16.25%
Σ	80	100%

Table 3.6 Accumulated relative frequency (F_r)

Value °C	f_a	f_r	F_r
18	15	18.75%	18.75%
19	18	22.5%	41.25%
20	12	15%	56.25%
21	22	27.5%	83.75%
22	13	16.25%	100%

$$f_r = \frac{fa}{n}$$

Table 3.5 shows that the values of 19 and 21 °C together represent 50% of the values. The remaining 50% are distributed by 18, 20, and 22 °C.

- *Accumulated relative frequency (F_r)*—corresponds to the sum of the relative frequencies of the current class with the sum of the immediately preceding relative frequencies (Table 3.6).

  More than half of the data (56.25%) is between 18 and 20 °C!

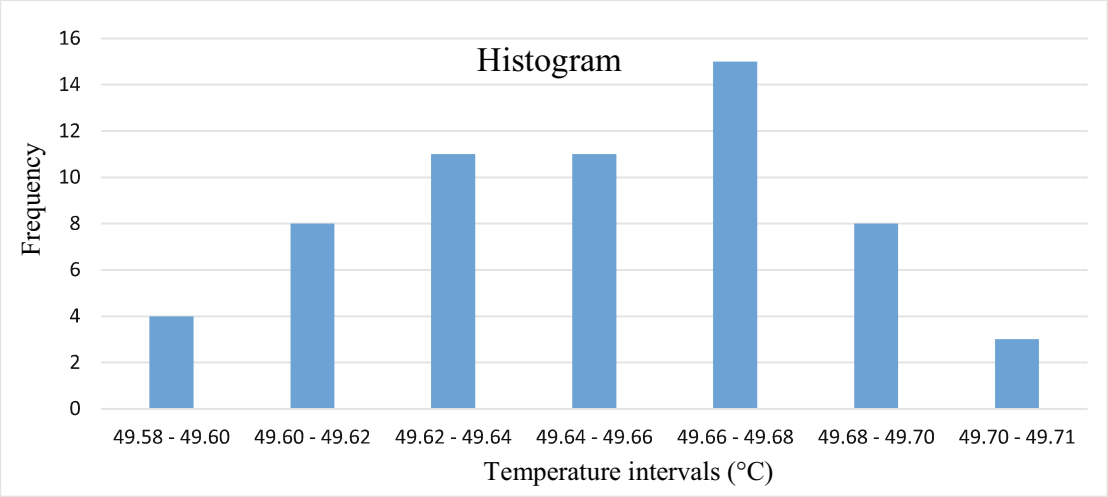
3.2.2.3 Histogram

A histogram is a bar graph that shows a frequency distribution, that is, the table where we present the data collected due to the frequency of its occurrence. The base of a rectangle represents a class of the frequency table. The height of the bar is proportional to the frequency value contained in the class. The horizontal scale of the graph is quantitative. The vertical scale indicates the absolute or relative frequency.

This data must be divided into class intervals. An efficient method for determining the number of class intervals consists of obtaining the square root of the number

Table 3.8 Class interval and the frequency

Class interval	Frequency
49.58 °C ≤ x < 49.60 °C	4
49.60 °C ≤ x < 49.62 °C	8
49.62 °C ≤ x < 49.64 °C	11
49.64 °C ≤ x < 49.66 °C	11
49.66 °C ≤ x < 49.68 °C	15
49.68 °C ≤ x < 49.70 °C	8
49.70 °C ≤ x < 49.71 °C	3



Graph 3.1 Stabilized oven temperature distribution histogram

The same Excel also automatically makes the histogram by selecting the data and clicking on the histogram in the data tab within the data analysis icon (Graph 3.1).

Note that the mean oven temperature is worth 49.64 °C and is in the middle of the frequency distribution in the histogram.

This is a feature of most statistical distributions, as we will see later in this chapter.

3.2.3 Probability Density Function (PDF)

One probability density function $f(x)$ describes the distribution of a random variable x. If a value x is very likely to occur within an interval [a, b], its PDF $f(x)$ will be significant in this interval. The PDF $f(x)$ then describes the probabilities associated with a random variable.

The PDF $f(x)$ may be discreet or continuous, depending on whether the variable x is discrete or continuous.

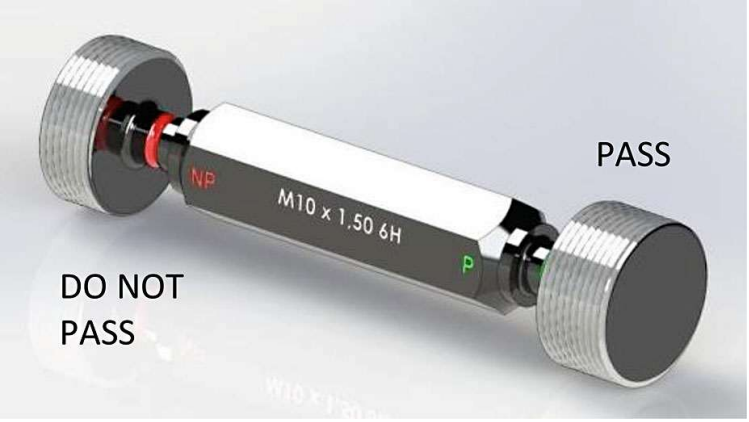


Fig. 3.3 Standard caliber. (https://www.calibratools.com.br)

3.2.3.1 Discrete Density Function

A discrete PDF describes the behavior of the variables that provide integer and finite values.

Example 1: It is very common to use measurement devices called calibers, which inform whether one piece conforms or not. Variable X then has only two values: pass/do not pass or go/do not go (Fig. 3.3).

The discrete PDF that characterizes the example well is the *Binomial Distribution*.

A binomial distribution is adequate when the results of a random variable are grouped into only two classes or categories. These categories must be mutually exclusive. For example, a manufactured product may be perfect or defective, an answer may be correct or wrong, and a telephone call is local or long distance.

Even continuous variables can be divided into two categories. For example, a car's speed may be below or above the legal limit on a road. These categories are usually called *success* or *failure*.

Application conditions:

- They are made in independent repetitions of the experiment, that is, the result of repetition is not influenced by others.
- The probability of success p and failure $(1 - p)$ remain constant in all repetitions.

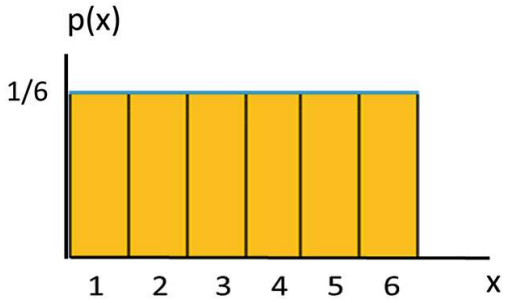
The following expression gives the binomial model:

$$P(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad x = 0, 1, \ldots, n \tag{3.5}$$

$$\text{Mean} \rightarrow \mu = np \tag{3.6}$$

$$\text{Standard deviation} \rightarrow \sigma = \sqrt{np(1-p)} \tag{3.7}$$

The binomial model is usually used in quality control when sampling a large population. In these applications, x represents the number of defective observations in a n-sample.



Graph 3.2 Discreet uniform PDF

3.2.3.2 Continuous Density Function

For a continuous random variable x, a PDF $f(x)$ is such a function that:

$$f(x) \geq 0$$

$$f(x) = \int_{-\infty}^{+\infty} f(x)\, dx = 1$$

$$P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$$

In Graph 3.3, the area under $f(x)$ represents the probability of x assuming a value between a and b (Graph 3.3).

$$\mu = \frac{\sum_1^k x_i}{k} = 3,5 \qquad \sigma = \sqrt{\frac{\sum_1^k (x_i - \mu)^2}{k}} = \sqrt{\frac{\sum_1^k (x_i - 3,5)^2}{6}} = 1,71$$

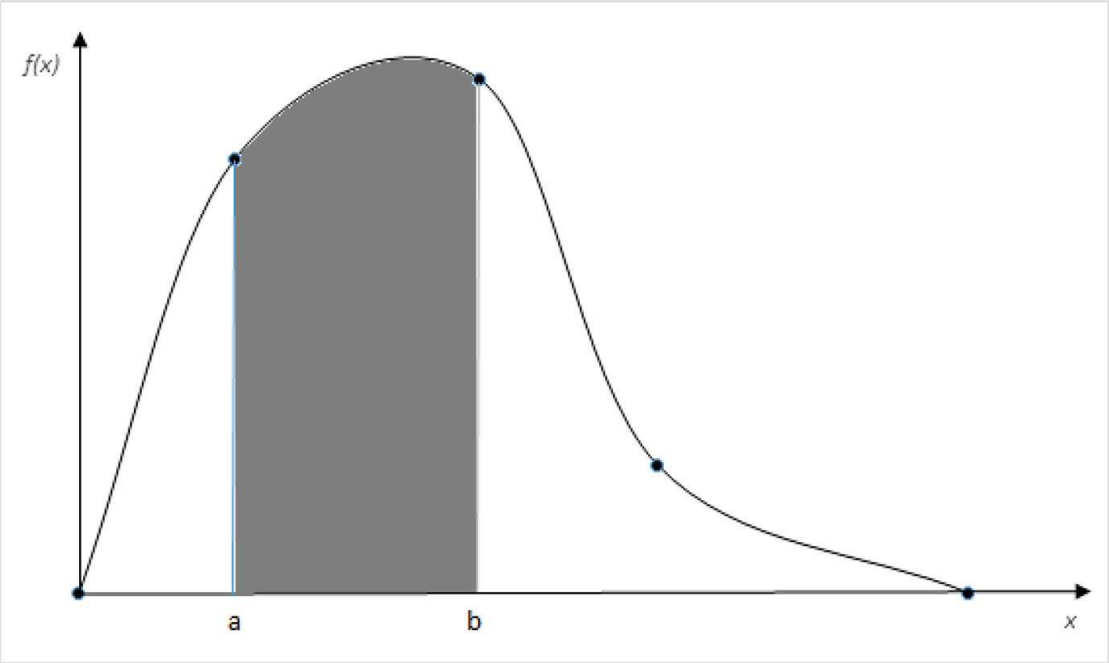
For a discrete random variable X, the sum of the discrete distribution $P(x)$ values between the boundaries $-\infty$ and $+\infty$ always results in one.

Remembering the example of the dice, the sum of $P(x)$ will be:

$$P(-\infty \leq X \leq +\infty) = P(1 \leq X \leq 6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$$

3.2.4 Mean and Standard Deviation of a Probability Distribution

Probability distributions are characterized by their mean value and their standard deviation. When the mean in question is the population's mean, we designate the Greek letter μ. When it comes to the mean of a sample, we designate by $\bar{x}$.



Graph 3.3 Probability $P\ (a \leq X \leq b)$

The mean (μ), or expected value $E(X)$, is the best estimate of a measurement and is defined by the equation:

$$\mu = E(X) = \int x.f(x)dx \tag{3.13}$$

where X is a random variable, and $f(x)$ is a PDF.

Although the mean sample or the mean population is valid, it is important to know how dispersed the data around the mean is. The variable that measures the dispersion of these data around the mean is called *standard deviation*, and its square is known as *variance*.

The variance of X is denoted by σ^2 or $V(X)$ and defined by the expression:

$$\sigma^2 = V(X) = E\left(((x - E(x))^2\right) = \int (\mu - x)^2 f(x)dx \tag{3.14}$$

3.2.5 Distributions of Probabilities More Adopted in Metrology

We mention, in Sect. 3.2.1, that the most commonly used probability distributions in metrology are uniform or rectangular, triangular, normal or Gaussian, and Student's t.

Now, let us study the main features of these distributions and their applications in metrology.

3.2.5.1 Rectangular or Uniform Distribution

We will face a uniform or rectangular distribution when the probability distribution is constant in a defined interval (Graph 3.4).

$$f(x) = \frac{1}{b-a}; \qquad a \le x \le b \tag{3.15}$$

$$f(x) = 0 \qquad x < a \ \text{ ou } \ x > b.$$

The mean of a uniform continuous random variable x is defined by Eq. (3.13): $\mu = E(X) = \int x.f(x)dx$.

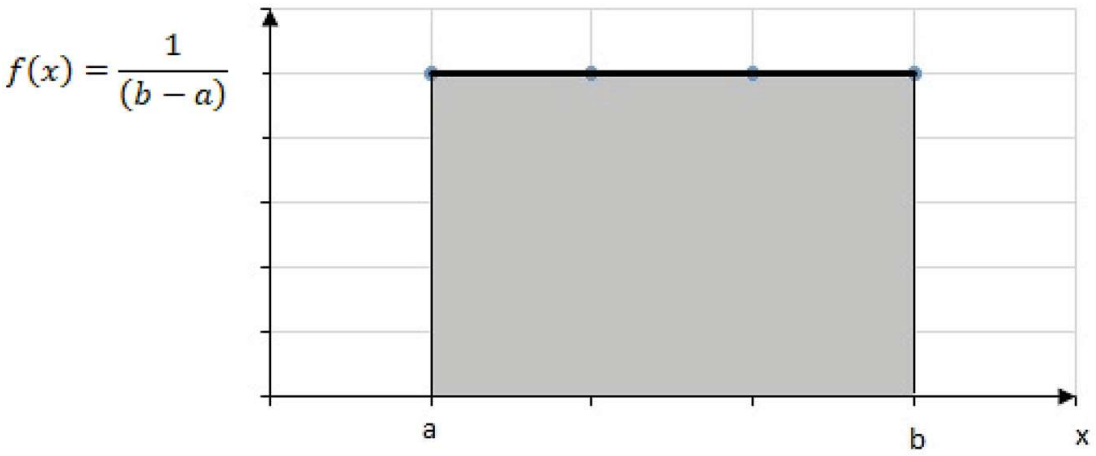
Integrating within the limits between a and b and adopting $f(x) = \frac{1}{(b-a)}$, we have:

$$\int_a^b \frac{x}{b-a}dx = \frac{x^2}{2(b-a)}\Bigg|_a^b$$

$$\mu = \bar{x} = \frac{a+b}{2} \tag{3.16}$$

Using the variance definition by Eq. (3.14), we have:

$$\sigma^2 = \int (\mu - x)^2 f(x)dx$$



Graph 3.4 Uniform or rectangular distribution

3.2.5.2 Symmetrical Triangular Distribution

When the probability distribution is more prominent in the central part, at a defined interval, and decays linearly at the ends, we will face a triangular distribution.

In many cases, it is more realistic to expect values near the boundaries to be less likely than those near the midpoint. Replacing the rectangular distribution with a symmetrical triangular distribution is reasonable (Graph 3.7).

$$f(x) = \begin{cases} 0 & \text{para } x < a \\ \dfrac{4(x-a)}{(b-a)^2} & \text{para } a \le x \le \dfrac{a+b}{2} \\ \dfrac{4(b-x)}{(b-a)^2} & \text{para } \dfrac{a+b}{2} \le x \le b \\ 0 & \text{para } x > b \end{cases} \tag{3.19}$$

For triangular distribution with the mean $\bar{x}$ at the center of interval a, b, we have:

$$\mu = \bar{x} = \frac{a+b}{2} \tag{3.20}$$

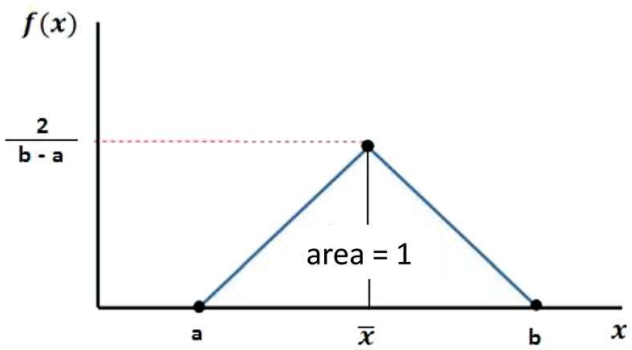
and the standard deviation is:

$$\sigma(x) = s(x) = \frac{b-a}{\sqrt{24}} = \frac{b-a}{2\sqrt{6}} \tag{3.21}$$

We adopted the expression $s(x)$ for the sample standard deviation and the expression $\sigma(x)$ for the population standard deviation. The same equation gives the standard deviation in a triangular distribution.

Solved Exercise 3.4

Suppose we calibrate a pressure gauge with a measurement interval (0 to 40) bar and a resolution of 1 bar using a comparative pump and fix the calibration points on the object gauge at 10 bar, 20 bar, 30 bar, and 40 bar (Fig. 3.4).



Graph 3.7 Symmetrical triangular distribution

3.2.5.3 Asymmetrical Triangular Distribution

In an asymmetrical triangular distribution, the value with the highest probability is the mode of the dataset, which is not equal to the average. So, when the probability distribution is higher in *mode*, in a defined interval, and decays linearly at the ends, we will be facing an asymmetrical triangular distribution (Graph 3.9).

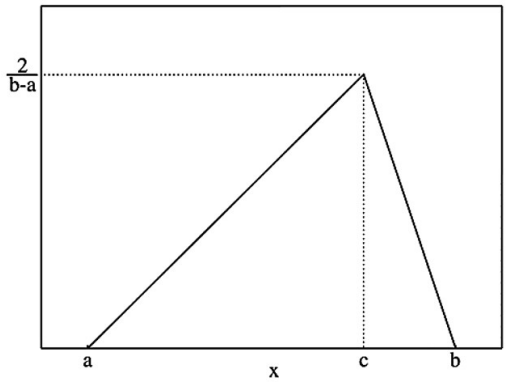
$$f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(c-a)} & a \le x < c \\ \dfrac{2}{b-a} & x = c = (\text{mode}) \\ \dfrac{2(b-x)}{(b-a)(b-c)} & c < x \le b \\ 0 & x < a \;\; e \;\; x > b \end{cases} \tag{3.22}$$

The mean and standard deviation expressions for this distribution are:

$$\mu = \frac{a+b+c}{3} \tag{3.23}$$

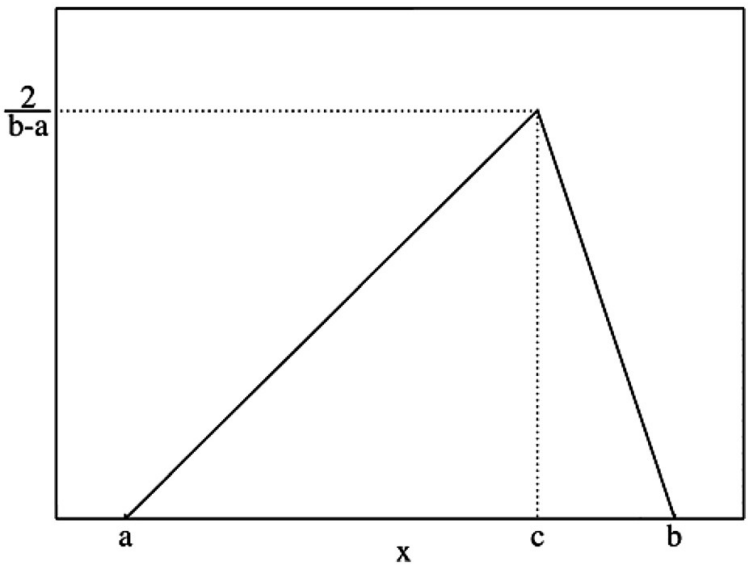
$$\sigma = \sqrt{\frac{a^2+b^2+c^2-ab-ac-bc}{18}} \tag{3.24}$$

The probabilities accumulated for this distribution are:



$$f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(c-a)} & a \le x < c \\ \dfrac{2}{b-a} & x = c = (mode) \\ \dfrac{2(b-x)}{(b-a)(b-c)} & c < x \le b \\ 0 & x < a \; e \; x > b \end{cases}$$

Graph 3.9 Asymmetrical triangular distribution



Graph 3.10 Asymmetrical triangular distribution of Solved Exercise 3.6. $a = 10$ °C. $b = 30$ °C. $c = 25$ °C

$$\mathrm{P}(x) = \begin{cases} \dfrac{(x-a)^2}{(b-c)\,(c-a)} & a \le x < c \\ \dfrac{c-a}{b-a} & x = c \\ 1 - \dfrac{(b-x)^2}{(b-a)\,(b-c)} & c < x \le b \end{cases} \tag{3.25}$$

Solved Exercise 3.6
Consider that in a set of measurements, the lowest value found was 10 °C, the largest was 30 °C, and the mode was 25 °C. What is the probability that a new measurement is less than 20 °C? What is the probability of being greater than 26 °C?

Solution (Graph 3.10)

$$P(x<20) = \frac{(x-a)^2}{(b-a)(c-a)} = \frac{(20-10)^2}{(30-10)(25-10)} = 0.33 \rightarrow 33\%$$

$$P(x>26) = 1 - \frac{(b-x)^2}{(b-a)(b-c)} = 1 - \frac{(30-26)^2}{(30-10)(30-25)} = 0.84 \rightarrow 84\%$$

3.2.5.4 Normal or Gaussian Distribution

Normal or Gaussian distribution is undoubtedly the most important PDF. Several variables behave according to a Gaussian distribution.

3.2.5.5 *Student's T* Distribution

When we perform a small number of measurements, less than 30, we realize that even when the sample belongs to a normal distribution, its histogram does not take the shape of a bell, typical of this distribution.

To visualize this feature, we randomly generate in Excel 1000 values belonging to a normal distribution of population average $\mu = 2.00$ and standard deviation of the population $\sigma = 0.40$.

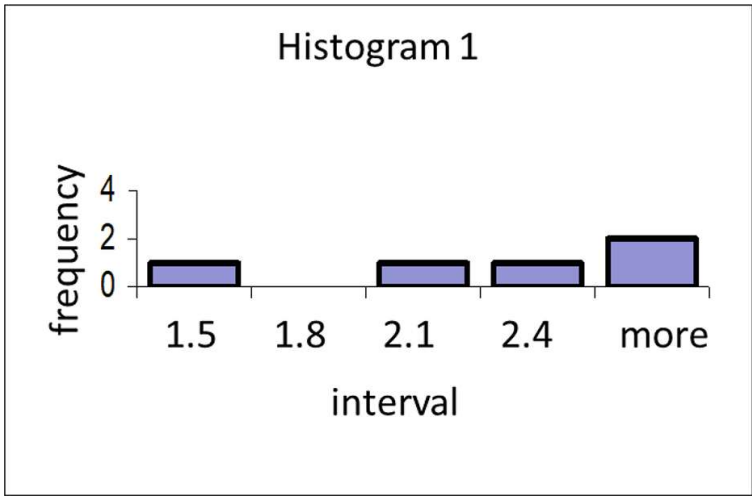
From this population (let us consider that the data generated are large enough to be viewed as the population), we remove sample sizes $n = 5$, $n = 20$, $n = 100$, and $n = 1000$. Our goal is to build histograms of different samples and observe their behavior.

3.2.5.6 Sample Analysis with $N = 5$ (Graph 3.17)

The result found was $\overline{x} = 2.09$ and $s = 0.43$

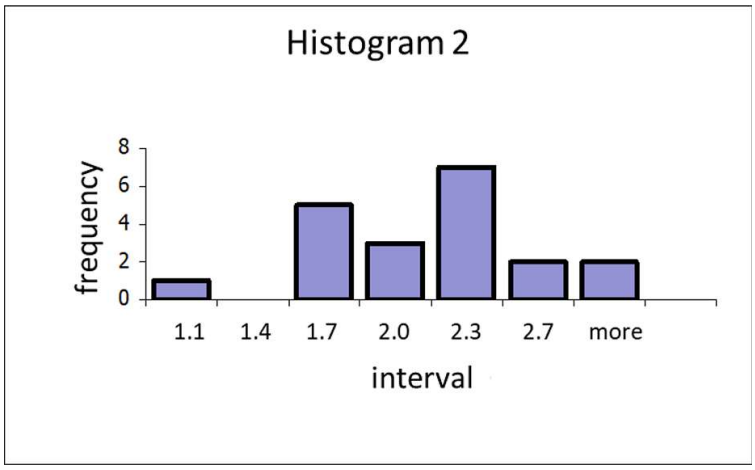
Interval	Frequency
1.5	1
1.8	0
2.1	1
2.4	1
More	2

Note that although the values are removed from a normal distribution with a mean of 2.00 and a standard deviation of 0.40, the mean of the five values is worth 2.09 and a standard deviation of 0.43. This is because we only get a mean of 2.00 and a deviation of 0.40, when we have all the values that generated the normal curve (infinite values).



Graph 3.17 Data and histogram to $n = 5$

Graph 3.18 Data and histogram to $n = 20$



Another essential feature is that the histogram containing the five values does not look like a normal distribution (bell form). This will only happen as the sample number approaches the population number.

3.2.5.7 Sample Analysis with $N = 20$ (Graph 3.18)

The result found was $\bar{x} = 1.97$ and $s = 0.44$

Interval	Frequency
1.1	1
1.4	0
1.7	5
2.0	3
2.3	7
2.7	2
More	2

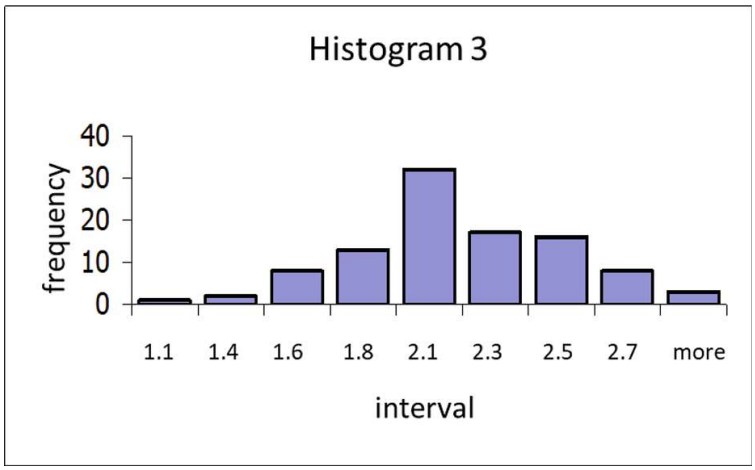
3.2.5.8 Sample Analysis with $N = 100$ (Graph 3.19)

The result found was $\bar{x} = 2.05$ and $s = 0.36$

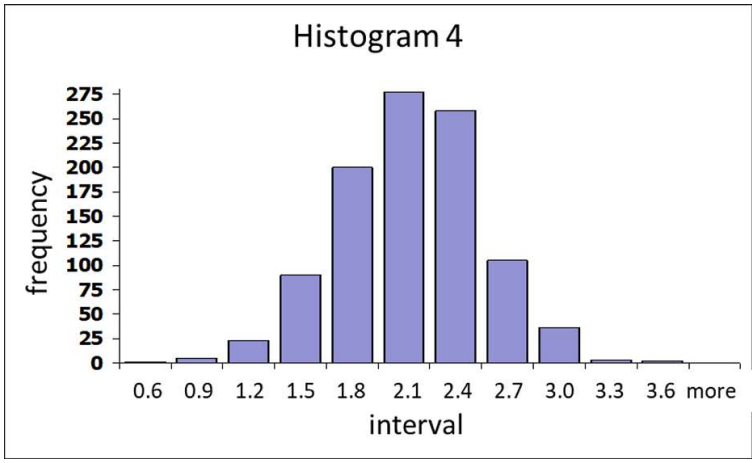
Interval	Frequency
1.1	1
1.4	2
1.6	8
1.8	13
2.1	32
2.3	17

(continued)

Graph 3.19 Data and histogram to $n = 100$



Graph 3.20 Data and histogram to $n = 1000$



Interval	Frequency
2.5	16
2.7	8
More	3

3.2.5.9 Sample Analysis with $N = 1000$ (Graph 3.20)

The result found was $\bar{x} = 2.00$ and $s = 0.40$

Interval	Frequency
0.6	1
0.9	5
1.2	23
1.5	90
1.8	200
2.1	277

(continued)

3.3 Proposed Exercises

3.3.1 Round correctly to one decimal digit.

(a) 34.450 m
(b) 23.852 m
(c) 8.351 m
(d) 19.7489 m
(e) 43.4501 m
(f) 43.852 m
(g) 52.3511 m
(h) 66.7205 m.

3.3.2 Check the number of significant digits in the following measurements:

(a) 1.320 m
(b) 0.050 kg
(c) 0.0001 km
(d) 9642 m^2.

3.3.3 Round correctly to three significant digits.

(a) 478.9 m
(b) 642.5 kg
(c) 123.4 L
(d) 56.150 cm.

3.3.4 Perform the following operations and present the result with the number of correct significant digits:

(a) 52.69 m + 36.8 m
(b) 68.487 m × 0.12 m

$$\sqrt{47.8\ \text{m}^2} - 1.36\ \text{m}$$

3.3.5 Round for one significant digit.

(a) 3682
(b) 0.00245
(c) 0.00058763
(d) 0.000030456.